

# KAAN BOREKCI

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## EDUCATION

### Northwestern University

*Master of Science in Mechanical Engineering* | GPA: 3.9/4.0

*Bachelor of Science in Biomedical Engineering* | Major GPA: 3.7/4.0 | GPA: 3.5/4.0 | **Dean's List**

**Evanston, IL**

*Expected Jun 2026*

*Jun 2025*

## WORK EXPERIENCE

### Shirley Ryan Ability Lab, Center for Bionic Medicine | Arun Jayaraman Lab

*R&D Engineer | Biomedical Machine Learning*

**Chicago, IL**

*May 2024 – Present*

- Pioneered a pipeline to assess 8+ MMPose 3D pose lifting models by processing 17 uncalibrated infant videos
- Increased feature generation by 50% through depth-based 2D-to-3D reconstruction of pseudo-ground truth sets
- Engineered an evaluation framework applying 10+ metrics across 17 keypoints per frame, validating predictions with clinically supported infant anatomical constraints (0.5, 1, and 6 months of age)
- Applied custom algorithm on 375k+ pseudo-ground truth frames with CUDA GPUs, identifying highest performing pose lifter for ML feature extraction, to enable diagnosis of infant motor delays up to 12 months earlier

### International Institute for Nanotechnology | Chad Mirkin Lab

*Nanotechnology Intern*

**Evanston, IL**

*Jun 2023 – Jul 2023*

- Optimized crystallization conditions of DNA-coated colloidal nanoparticles, reducing formation time by 48+ hrs
- Introduced novel ImageJ workflow analyzing 80+ hrs of crystal data, demonstrating results to Dr. Mirkin and lab

### Innovation and New Ventures Office at Northwestern | Dr. Lisa Dhar

*Technology Transfer Intern*

**Evanston, IL**

*Jul 2022 – Aug 2022*

- Built internal database tracking \$2.75B raised across 180+ startups, supporting tech-transfer strategy development
- Consolidated 12 yrs of scattered funding data into a guide used by the Associate VP of INVO to drive IP strategies

### Querrey Simpson Institute for Bioelectronics | John Rogers Lab

*Bioelectronics Research Intern*

**Evanston, IL**

*May 2020 – Sep 2020; Jul 2019 – Aug 2019*

- Evaluated embedded sensors for post-graft skin flap monitoring and hydrocephalus shunt flow detection
- Spearheaded experiments on phantom models, simulating physiological flow conditions to test device accuracy
- Coauthored peer-reviewed [Nature](#) article with experimental contributions and figure designs in Adobe Illustrator

## PROJECT EXPERIENCE

### Diaphragm Control for Pediatric Congenital Central Hypoventilation Syndrome

*R&D Engineer | Senior Capstone & Independent Project | Lurie Children's Hospital CAMP Division*

**Chicago, IL**

*Sep 2024 – Present*

- Selected to present in BMES 2025 (San Diego), patent pending with IRB-approved trials and IEEE paper underway
- Led 4-person team pioneering first-step platform towards a closed-loop diaphragm pacer regulation model
- Programmed embedded C firmware on RP2040, integrating multicore processing and communication protocols (I2C, UART, SPI) to handle IMU and CO2 sensor inputs with SD-card logging to synchronize capture at 100 Hz
- Guided team decision to implement backup through-hole Printed Circuit Board (PCB), reducing power usage by 20+ hrs and avoiding 15+ weeks of delays caused by failed high-fidelity iterations
- Partnered with clinicians to define clinically relevant features and applied signal processing algorithms on IMU and CO2 inputs, validating device feasibility while testing early feedback models

### Continuous Live Web Streaming Device

*Course Project for ELEC\_ENG 326: Electronic System Design*

**Evanston, IL**

*Jan 2025 – Mar 2025*

- Collaborated in a 2-person team to present a full-stack prototype within 10 weeks, with custom Eagle PCB, ESP32 web streaming, and 3D-printed enclosure
- Developed SAM4S MCU firmware in C integrating camera and ESP32 via SPI/UART, enabling 320x240 live stream

## SKILLS

- **Programming:** Python, Embedded C/C++, MATLAB, HTML, CSS, Git, MMPose, OpenCV, NumPy, scikit-learn, PyTorch
- **Tools:** Ansys, Fluent, EAGLE, KiCAD, Illustrator, Photoshop, Premiere Pro, CAD, OnShape, Solidworks, ImageJ
- **Hardware & Prototyping:** Raspberry Pi, Arduino, STM32, ESP32, PIC32, Microchip, nRF, PCB, Signal Processing
- **Languages & Interests:** Turkish (Native), English (Native), German (A2), Volleyball (HS Captain), Golf, Skiing, Tennis